

Aug. '26

TMSMultiLab Hackathon '26



<https://github.com/TMSMultiLab/TMSMultiLab/wiki/Hackathon2026>

Hackathon Introduction

Outline:

- 1. TMSMultiLab**
- 2. Aims**
- 3. Sessions**
- 4. Topics**
- 5. Teams**

Hackathon Introduction

TMSMultiLab

- Began in ~[2023](#)
- ~100 members (~20 countries, >50 institutions)
- Monthly meetings (last Friday, 14h UK time)
- Workshop in April 2025 ([TMS@40](#))
- First (annual?!) **Hackathon**, August 2026

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AIMS

- Work together to develop the repository
- Summarise the 'state of the art'
- Identify issues, best practice, gaps
- Curate shared, open dataset
- Write code to do all of the above

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SESSIONS

- **Github** (yesterday)
- **Introduction** (now)
- **Day 1** (morning, afternoon, ~5 hours)
- **KeyNote** (Today 16h Paris, 08h Denver)
- **Day 2** (morning, afternoon, ~5 hours)

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TOPICS

- A) EMG / MEP Onsets and Offsets**
- B) MEP timing and variability**
- C) BOTH: Develop a shared open dataset**

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TEAMS*

A1)  Hedgehogs (Europe, Topic A)

B1)  Squirrels (Europe, Topic B)

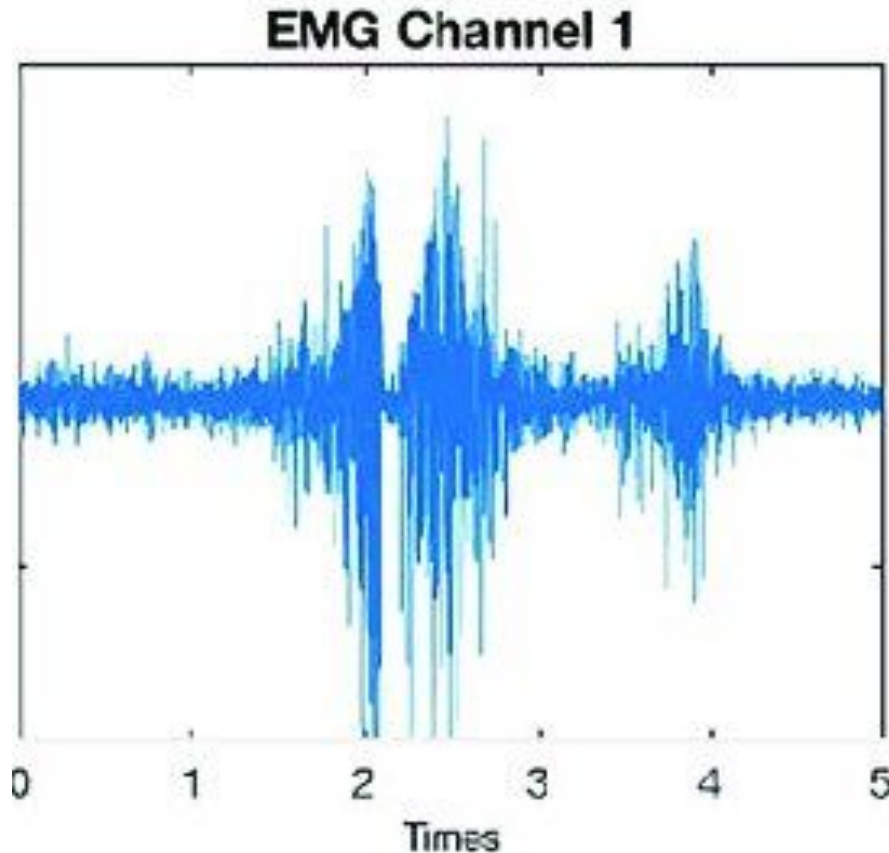
A2)  Wolves (Americas, Topic A)

B2)  Rattlesnakes (Americas, Topic B)

* assuming that everyone comes along...

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TOPIC A: EMG / MEP Onsets and Offsets

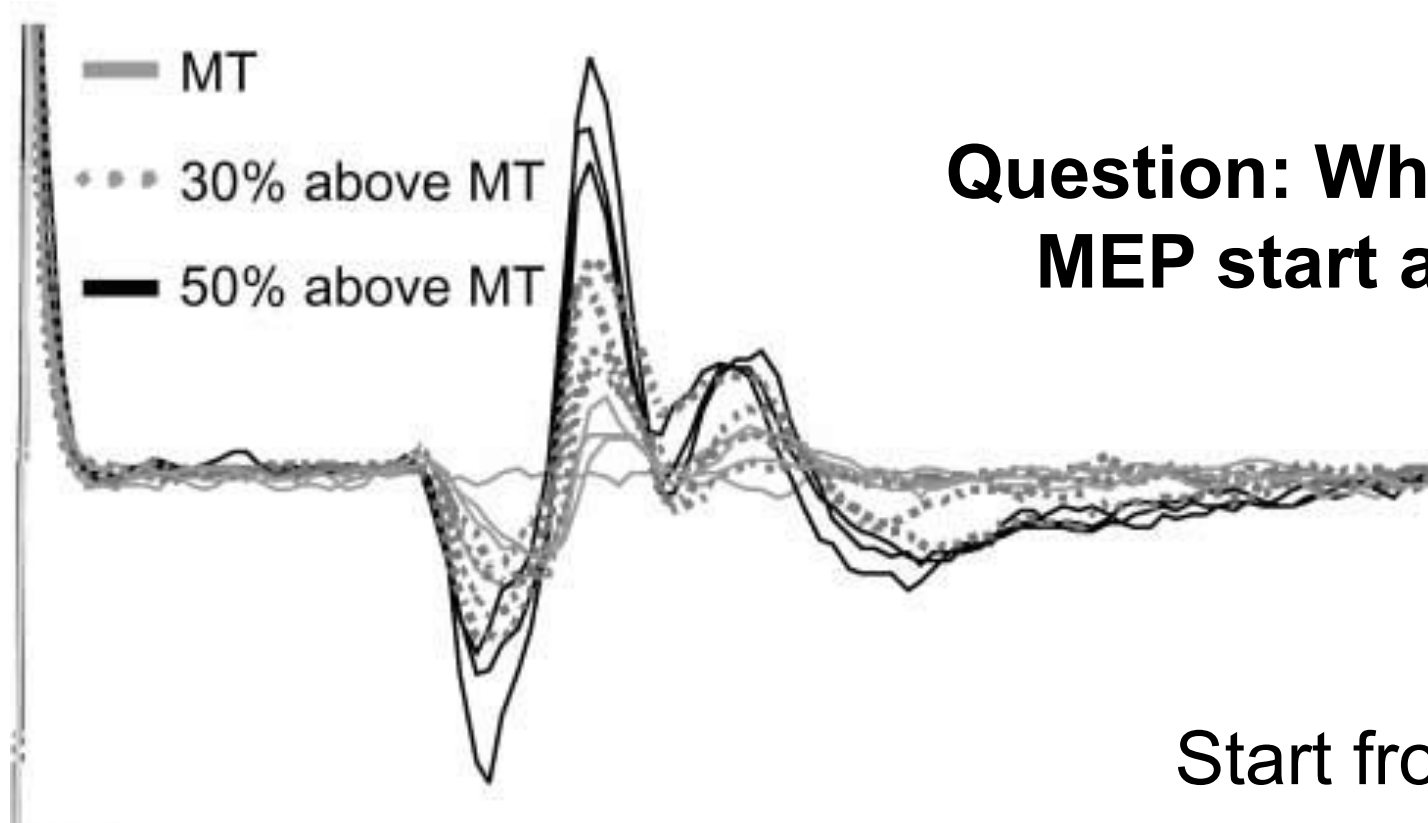


Question: When does the EMG start & stop?

Start from [[EMG wiki](#)]

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TOPIC A: EMG / MEP Onsets and Offsets

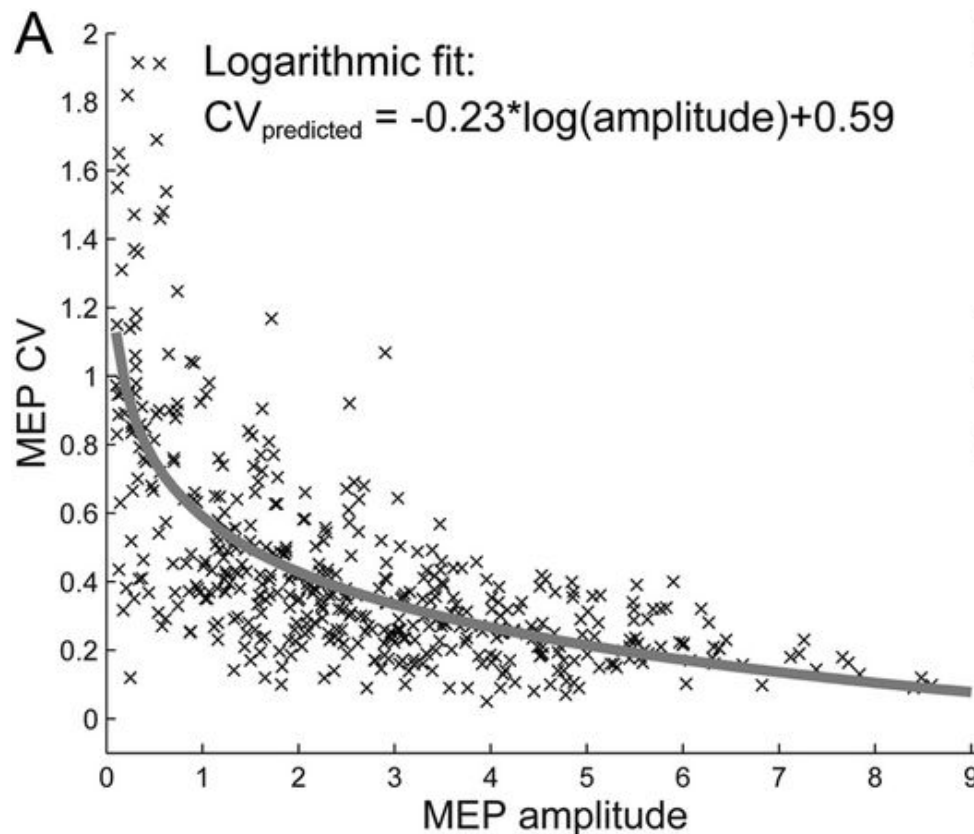


**Question: When does the
MEP start and stop?**

Start from [[MEP wiki](#)]

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TOPIC B: MEP variability and timing



Question: How does the precise *timing* of MEPs affect their variability?

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TOPIC C: Construct an shared, open dataset

- **We need a “Gold-Standard” set of test data**
- **It must be...**
 - **BIG!** (at least 100,000 MEPs / EMG epochs)
 - Open & free to redistribute (ideal: already online)
 - Diverse sample: healthy & clinical; young and old
 - Diverse tasks: rest, contraction, movement
 - Diverse muscles: hand, arm, head, trunk, legs, feet
 - Diverse methods: hardware, sampling, filters

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ON TOPICS A & B...

- **You are not the first to work on these topics!**
- **So...**
 - Find the best-available & latest existing work
 - Describe & document it on the wiki
 - Compare & contrast methods
 - Find open datasets and code
 - Try the code, document whether it works
 - Develop & test new code...

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HOW TO WORK TOGETHER?

- **Communicate on Zoom & Slack**
- **Assign tasks & try not to overlap. e.g.:**
 - 1: Coordinator
 - 2: Organises the wiki page(s)
 - 3: Literature searching
 - 4: Data curation
 - 5: Code testing
 - 6: Code development

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GUIDELINES & STANDARDS

- Acknowledge & attribute work (references, links)
- Minimise bias (be systematic, find all sources if you can)
- Contributorship
 - Sign your main bits of work (e.g., data / code)
 - If you used LLMs: *declare this & check accuracy*
 - Wherever possible: two humans check
- Standards
 - Data: aim for [BIDS-compatible](#) organisation
 - Reporting/metadata: [TMS-RAT](#) checklist

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DISCUSSION

